**Templated Problem Demo** Algebra 2 / Calculus Version A

Name: ________________________________________

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**1. (12 pts)** Let $f(x) = (x+3)(x-1)(x-4)$.

a. Find all real zeros of $f$. **Answer:** $x = -3, 1, 4$

b. Determine the following key features:

**i.** The y-intercept. **Answer:** $f(0) = 12$ **ii.** Where $f(x) > 0$. **Answer:** $x \in (-3, 1) \cup (4, \infty)$

c. Describe the end behavior of $f$:
i. As $x \to -\infty$: **Answer:** $f(x) \to -\infty$
ii. As $x \to +\infty$: **Answer:** $f(x) \to +\infty$

**2. (10 pts)** The area of a triangle with sides $a$ and $b$ and included angle $\theta$ is $A = \frac{1}{2}ab\sin(\theta)$.

1. If $a = 2$, $b = 3$, and $\theta$ is increasing at $\frac{d\theta}{dt} = 0.2$ rad/s, find a formula for $\frac{dA}{dt}$ in terms of $\theta$.

   **Answer:** $\frac{dA}{dt} = \frac{3\cos(\theta)}{5}$

2. Find the rate at which the area is changing when $\theta = \pi/3$.

   **Answer:** $\frac{dA}{dt} = \frac{3}{10}\text{ft}^2$ / s